each other, and it is essential for the data scientists and machine learners to understand the lineage in order to see which data is derived from where.

DEFINITION When referring to *lineage* or *data lineage*, we typically mean a representation of where the individual pieces of data came from. If you have a dataset that consists of revenues per customer, two base sources will likely be involved: the payments of some customer IDs, and then the customer dataset that maps IDs to other properties like name or gender. *Lineage* simply means making exactly this part of the information visible in any way. It means listing the two sources, the customer dataset, and the customer IDs—maybe with a tree-like representation, maybe with more metadata. It also means we need some kind of manual or, hopefully, automated way of recording this information.

So people do what they usually do and call up the two engineers on the data team and ask them to help them dig through the catalog. If the platform is successful and has lots of data, the central team will end up doing nothing else but answering calls all day long.

But that leaves them with no time to work on the platform, and it won't scale. So let's look at X as a service to understand how platform teams can improve their communication with others.

7.2.1 X as a service explained

Team Topologies is a framework created by Matthew Skelton and Manuel Pais for modern technology organizations. Described in their book *Team Topologies: Organizing Business and Technology Teams for Fast Flow* (IT Revolution, 2019), this framework helps to design teams and their interactions in a way that optimizes the flow of produced value.

As such, it is a great example of socio-technical architecture. In the framework, we're considering only three ways teams can interact, and one of them is X as a service.

We will give you a better definition of this in a minute, but we would like to return to the wood workshop example once more.

WORKSHOP TOOL-BUILDING AS A SERVICE